

ASHWINI BHARDWAJ

Jaipur, Rajasthan

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TECHNICAL SKILLS

Agentic AI & LLM Engineering: LLMs, Transformer Models (GPT, Llama, BERT), Agentic AI Workflows (Tool Use, Planning, Memory/State), RAG, LangChain, LangGraph, MCP, Prompt Engineering, Fine-Tuning (QLoRA/PEFT), Unsloth
LLM Evaluation & Observability: LangSmith, RAGAS, LLM-as-a-Judge, Prompt Guardrails
Backend & APIs: Python, FastAPI, Flask, RESTful APIs, JSON, SQL, MySQL, PostgreSQL
Cloud & Deployment: Docker, AWS, Git/GitHub, Hugging Face Hub, Databricks
Machine Learning & Deep Learning: Scikit-learn, TensorFlow, Keras, Pandas, Numpy, Supervised/Unsupervised Learning, Feature Engineering
Vector Search & Retrieval: Chroma, Qdrant, Embeddings, Semantic Search, Hybrid Search

PROJECTS

Prompt Safety Classifier (LLM Fine-Tuning) | PyTorch, Hugging Face, Transformers, Unsloth, PEFT (QLoRA) Jun 2026

- Curated a 150K+ prompt safety dataset from multiple open-source repositories into a unified 3-class (benign, harmful, jailbreak) benchmark, then fine-tuned Llama-3.2-1B-Instruct on a representative training subset using QLoRA (PEFT) and Unsloth.
- Built an end-to-end fine-tuning pipeline (preprocessing, tokenization, 4-bit NF4 quantization, LoRA adaptation, training, evaluation, inference), updating less than 1% of model parameters for memory-efficient training.
- Published the fine-tuned model on Hugging Face Hub and deployed an interactive Gradio application for real-time prompt safety classification, enabling reproducible inference.

Agentic Research & Content Generation Pipeline | Python, LangGraph, LangChain, Tavily, arXiv API Apr 2026

- Designed an agentic AI workflow in LangGraph integrating 2 retrieval tools (Tavily web search + arXiv academic API) with conditional planning logic to decide between direct and research-backed generation, improving factual grounding in long-form AI outputs.
- Implemented a 6-node stateful agent graph with tool orchestration and conditional routing, reducing redundant API calls by ~25%.
- Engineered parallel retrieval execution across multiple knowledge sources/tools, reducing average pipeline latency by ~35% compared to sequential fetching.

Natural Language Query Engine for SQL Databases | Python, LangChain, LangGraph, Flask, MySQL Nov 2025

- Built a natural language-to-SQL agentic system over a MySQL schema, exposed via a Flask RESTful API, enabling non-technical users to query structured databases using conversational inputs with no SQL knowledge.
- Designed a 5-stage LangGraph pipeline (intent parsing, SQL generation, validation, execution), achieving ~90% query execution success rate on test inputs.
- Implemented a query validation layer that reduced hallucinated or malformed SQL generation by ~45%, improving overall execution reliability.

EXPERIENCE

Envision Virtue

Dec 2024 – May 2025

Data Science Intern

Remote, India

- Trained and evaluated 5+ ML models (classification and regression) on real-world datasets of 50K+ records using Scikit-learn and TensorFlow, improving baseline model accuracy by ~12% through iterative tuning.
- Built data preprocessing and feature engineering pipelines that reduced missing value rate to under 1% and improved downstream model F1 scores by ~8%.
- Automated model generation workflows using Python and ML libraries, reducing manual preprocessing and training effort by 40% and cutting pipeline execution time from hours to minutes.

EDUCATION

VIT Bhopal University

Aug 2023 – Oct 2025

Master of Computer Applications | CGPA: 8.79

Bhopal, India

Birla Institute of Technology

Aug 2020 – May 2023

Bachelor of Computer Applications | CGPA: 8.62

Jaipur, India

CERTIFICATIONS

- AWS Educate: Introduction to Generative AI Amazon Web Services
- Software Engineer Job Simulation Accenture (Forage)
- Generative AI Fundamentals Databricks
- Databricks for Machine Learning Databricks